CSci 23: Module 1 Applications Development →

Emerging Technologies

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Course Outcome

Manage application development using version control.

Learning Outcome

Identify the fundamental concepts about emerging technologies.

Outline

1. Technology
2. Emerging Technologies

Emerging Technologies

Technology

What is Technology?

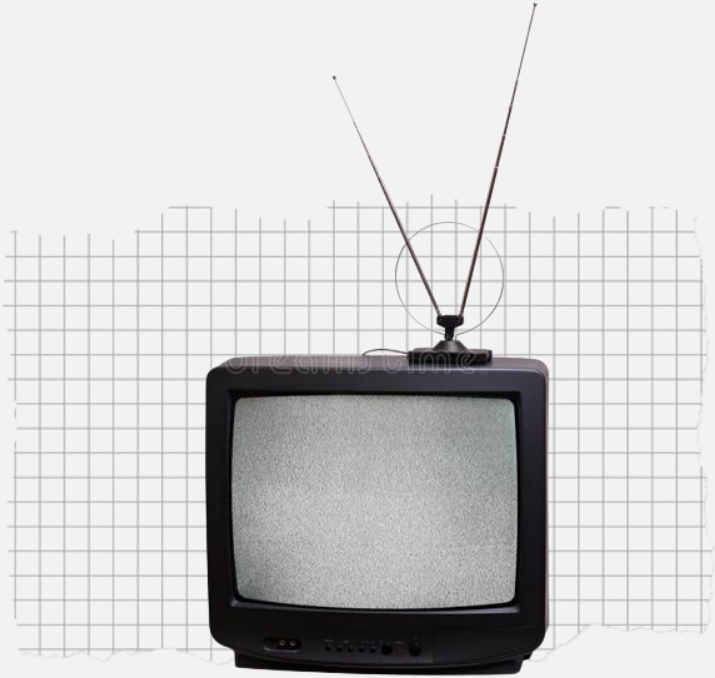
It is the practical application of scientific knowledge, tools, methods, and systems to solve problems, perform tasks, or achieve specific human goals.

1880s

1970s

1980s

1990s



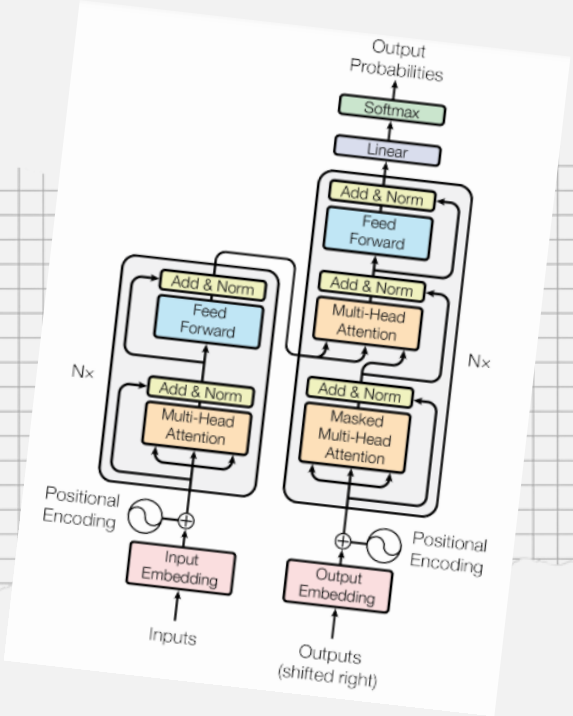
2000s

2007

cingular
9:41 AM
Text
Calendar
Photos
Camera
Calculator
Stocks
Maps
Weather
Notes
Clock
Settings
Phone
Mail
Safari
iPod

2009

2017



Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., & Polosukhin, I. (2023). Attention Is All You Need (No. arXiv:1706.03762). arXiv. https://doi.org/10.48550/arXiv.1706.03762

2030+

What is a Legacy Technology?

Any hardware, software, or application that is outdated, incompatible, or no longer supported by the developer or manufacturer.

Planned Obsolescence

A marketing strategy where products are designed to become outdated after a certain period, to drive sales and encourage consumers to buy the latest products.

What is Disruption?

It is the process of breaking apart and reorganizing existing systems, processes, or industries to create something new and better.

2022

1997

1998

2006

2008

2013

1994

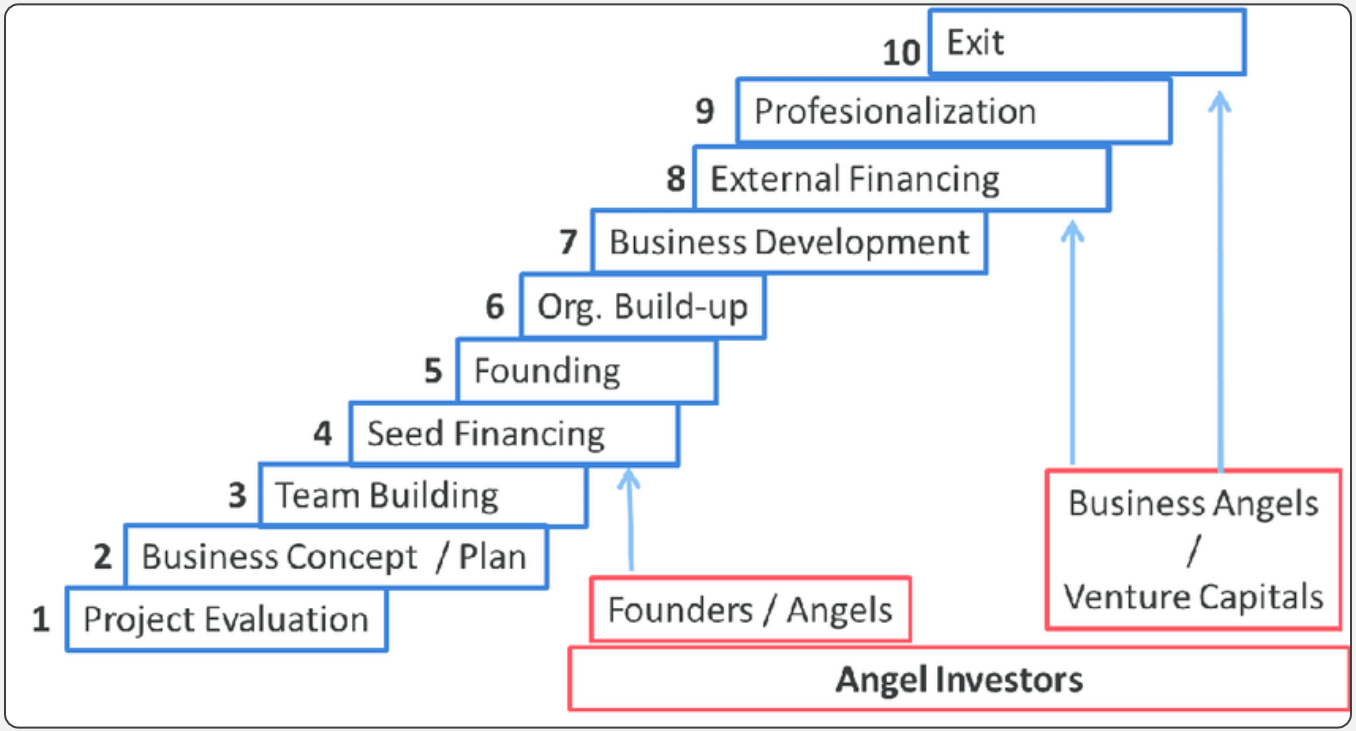
2004

2007

2009

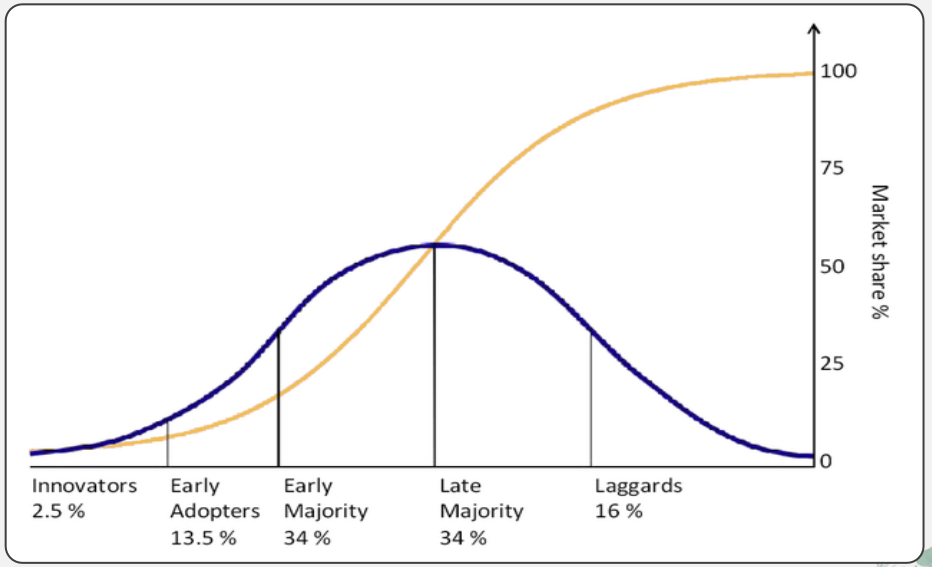
2015

What is Disruption?



What is Adoption?

It is the process of implementing and embracing a new product, process, or technology.



Emerging Technologies

Emerging Technologies

What is ‘Emerging’?

It describes something that is in the early stages of development or adoption, and is beginning to gain recognition, usage, or market acceptance.

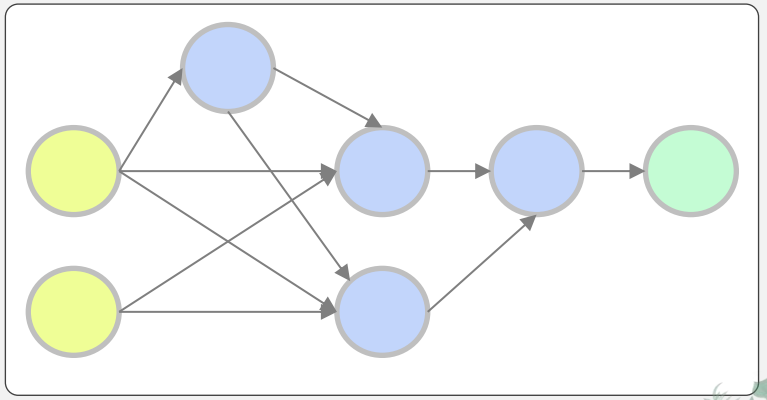
What is an Emerging Technology?

A technology that is still currently being developed and is not yet widely available or accepted but have the potential to have a significant impact on the way people live and work.

#1 AI Agents

It uses multi-modal generative models combined with autonomous decision-making, tool use, and goal-driven orchestration to perform tasks, coordinate actions, and operate across systems with minimal human intervention.

* It allows casual users to discover, build, and deploy mini-apps, without explicit coding.



https://developers.google.com/opal

#2 Enhanced AI/VR

It leverages both HW/SW technologies to enable immersive experience including navigation, scientific simulations, education and training, retail, and entertainment.

#3 Digital Twins

It focuses on creating logical representations of physical systems that integrate real-time operational data to monitor, analyze, and optimize assets, processes, and services.

#4 5G Expansion (and Early 6G)

It facilitates transformative technologies such as IoT, AR/VR, and autonomous vehicles by enabling real-time communication with high-speed and low-latency connections.

#5 Quantum Computing

It leverages quantum mechanics to process information exponentially faster while working alongside with the existing classical computers.

Summary

- A technology is the practical application of scientific knowledge, tools, methods, and systems.
- An emerging technology includes actively developed innovations that are not yet widely accepted but presents significant impacts in the society.
- A disruption occurs when a technology significantly impacts society, causing existing industries, processes, and systems to break or reorganize.

Scan me!

Evaluation Form

forms.gle/TncdC5nEKJZYucVz6

Thank You!

References

- Küfeoğlu, S. (2022). Emerging technologies for sustainable development: Innovation, value creation, and the SDGs. Springer International Publishing. https://doi.org/10.1007/978-3-031-07127-0
- https://www.startus-insights.com/innovators-guide/emerging-technologies-in-software/

Eneeergy Check!

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Zzz...

1

Interested 🤔

2

Motivated!!!